Activity 2: Constituency Parse Tree

Constituency Parse Trees

Constituency parsing is the process of analyzing the grammatical structure of a sentence by decomposing it into smaller, nested sub-phrases called constituents. Each constituent represents a meaningful grammatical unit, such as a Noun Phrase (NP), Verb Phrase (VP), or Prepositional Phrase (PP), which together illustrate how words combine to form larger linguistic structures. This hierarchical organization is typically represented as a parse tree, showing the relationships between components of the sentence. In Natural Language Processing (NLP), constituency parsing is fundamental because it enables intelligent systems to interpret sentence structure, identify syntactic roles, and derive meaning—critical steps in tasks like machine translation, question answering, and semantic analysis.

<u>Here are the parse trees for your three sentences:</u>

1. The government raised interest rates.

This sentence has a straightforward structure: a subject (NP) performs an action (V) on an object (NP).

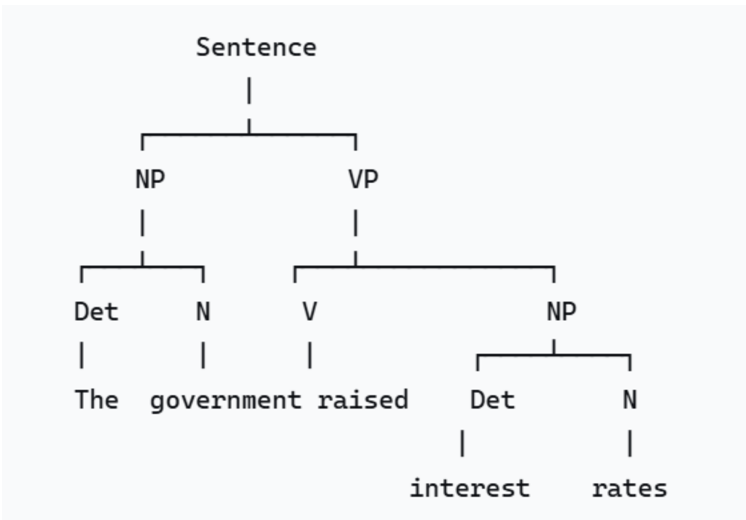

Figure 1: The government raised interest rates

2. The internet gives everyone a voice.

This sentence contains a ditransitive verb ("gives"), which means it has two objects: an indirect object ("everyone") and a direct object ("a voice").

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               Sentence
                  |
        ┌─────────┴─────────┐
        NP                  VP
        |                   |
   ┌────┴────┐         ┌────┴────┐
   Det       N         V         NP
   |         |         |      ┌───┴───┐
   The   internet    gives   Det      N
                              |       |
                              a     voice
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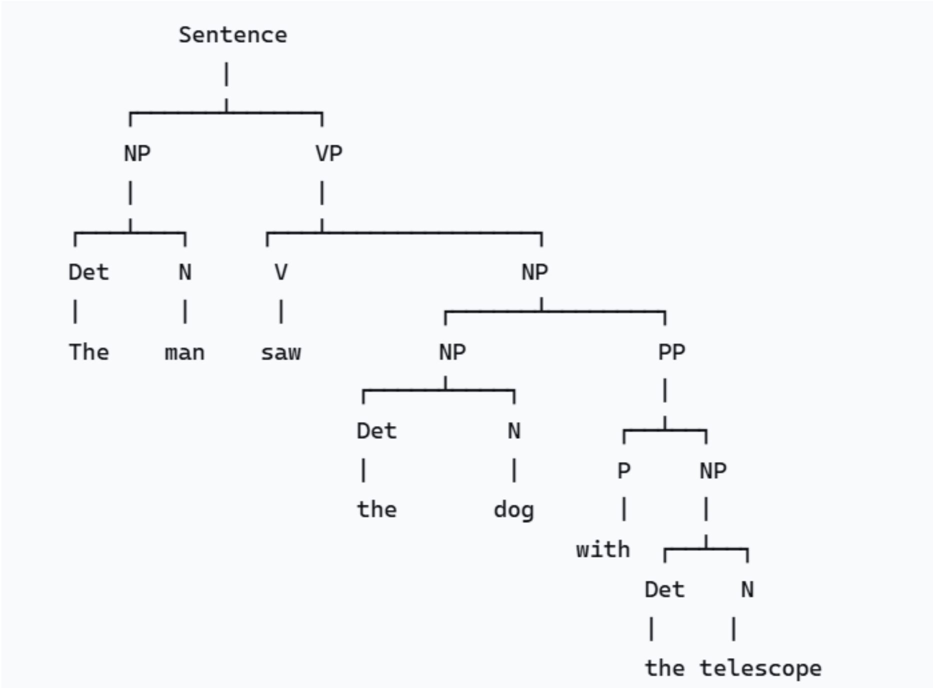
Figure 2: The internet gives everyone a voice.

3. The man saw the dog with the telescope.

This sentence is **structurally ambiguous**. The Prepositional Phrase (PP) "with the telescope" can modify either the verb "saw" (meaning the telescope was the instrument used for seeing) or the noun "dog" (meaning the dog had the telescope). This ambiguity is a classic challenge in NLP. An intelligent system must disambiguate this to understand the true meaning.

Interpretation A: The man used the telescope to see the dog.
(The PP "with the telescope" modifies the verb "saw")



Interpretation B: The man saw the dog that had the telescope.
(The PP "with the telescope" modifies the noun "dog")